

QA Evidence — Helix OTA

OpenDesign brand token package

Revision: 1 **Last modified:** 2026-07-09T00:00:00Z **Run ID:** 20260709-helix-brand-tokens **Scope:** author design-systems/helix-ota/ as a real OpenDesign-schema brand package (\$11.4.162 OpenDesign UI design-token system).

1. Schema source (cited)

OpenDesign was cloned READ-ONLY at .../scratchpad/open-design (github.com/nexu-io/open-design). The package was authored against these exact contract files in the clone:

Contract	File in clone	What it defines
Project manifest schema	design-systems/_schema/manifest.schema.ts	od-design-system-project/v1 — required top-level keys, files.* literal names, id slug rule
Token schema (TOKEN_SCHEMA)	packages/contracts/src/design-systems/token-schema.ts (re-exported by design-systems/_schema/tokens.schema.ts)	The 56 required custom properties across 4 layers (A1-identity / A1-structure / A2 / B-slot)
A2 fallbacks	design-systems/_schema/defaults.css	Sanctioned default values for A2 tokens (source for our non-canonical structural tokens)
Tailwind v4 renderer	packages/contracts/src/design-systems/derived-token-outputs.ts (renderTailwindV4Css)	Deterministic tailwind-v4.css derivation from tokens.css :root names

Contract	File in clone	What it defines
Guard	scripts/check-design-system-manifests.ts	Validates manifest shape, required files exist, and tailwind-v4.css is byte-exact with the renderer
Quality gate	scripts/check-design-system-package-quality.ts	Minimums for “migrated/rich” packages (see §4 note)

Reference example packages studied: design-systems/default, design-systems/github.

2. Palette provenance (token → source → value)

Primary canonical source: clients/ota-manager/src/index.css (the de-facto Helix HSL token set — shadcn/ui “blue” theme). Each HSL triple was converted to hex losslessly (colorsys, round-half-up). No Helix-OTA-specific logo/brand SVG exists in the repo, so no accent color was anchored to a logo (see §5).

Token	Source (index.css var)	HSL (light / dark)	Hex LIGHT	Hex DARK
--bg	--background	0 0% 100% / 222.2 84% 4.9%	#ffffff	#020817
--surface	--card	0 0% 100% / 222.2 84% 4.9%	#ffffff	#020817
--surface-warm	--secondary	210 40% 96.1% / 217.2 32.6% 17.5%	#f1f5f9	#1e293b
--fg	--foreground	222.2 84% 4.9% / 210 40% 98%	#020817	#f8fafc
--muted			#64748b	#94a3b8

Token	Source (index.css var)	HSL (light / dark)	Hex LIGHT	Hex DARK
	--muted- foreground	215.4 16.3% 46.9% / 215 20.2% 65.1%		
--border	--border	214.3 31.8% 91.4% / 217.2 32.6% 17.5%	#e2e8f0	#1e293b
--accent	--primary (brand blue)	221.2 83.2% 53.3% / 217.2 91.2% 59.8%	#2563eb	#3b82f6
--accent-on	--primary- foreground	210 40% 98% / 222.2 47.4% 11.2%	#f8fafc	#0f172a
--danger	-- destructive	0 72.2% 50.6% / 0 62.8% 30.6%	#dc2626	#7f1d1d
--radius-sm	--radius (0.5rem)	—	8px	8px

Non-canonical tokens (no value in index.css → sourced from the OpenDesign schema, NOT invented): --accent-hover / --accent-active — schema color-mix(...) formulas (_schema/defaults.css); adapt per theme via var(--accent). --success #16a34a, --warn #eab308 — schema A2 fallbacks. --fg-2 → var(--fg), --meta → var(--muted), --border-soft → var(--border) — B-slot aliases (ota-manager has only 2 text tiers / 1 border weight). - fonts (Tailwind default sans — ota-manager uses font-sans), type scale, spacing, section rhythm, radius md/lg/pill, elevation, focus, motion, container/leading/tracking — OpenDesign schema A2 defaults + default brand A1-structure scale (ota-manager relies on Tailwind defaults).

3. Validation result (ran the real schema + renderer)

Node v24 ran a standalone validator importing the clone's genuine parseDesignSystemProjectManifest, TOKEN_SCHEMA, isAllowedExtension, and renderTailwindV4Css. The full check-design-system-manifests.ts

guard was not run in-place because the clone is READ-ONLY; instead its exact validation logic (the same schema parser, the same tailwind renderer, the same `:root` regex) was invoked directly against the package. Output:

```
PASS manifest.json parses against od-design-system-project/v1
schema
PASS manifest id matches folder slug "helix-ota"
      guard-parsed :root blocks = 1; token names = 56
PASS tokens.css declares all 56 TOKEN_SCHEMA tokens in :root
PASS no unknown/unauthorized tokens in :root
      quality-gate token coverage: 56 >= 26 -> true
PASS tailwind-v4.css is byte-exact with
renderTailwindV4Css(tokens.css)
      DESIGN.md H2 sections = 9 (>= 7 -> true)
PASS DESIGN.md has 9 H2 sections
```

RESULT: ALL CHECKS PASS

Key point: the guard's `:root(?:\[])` regex parses exactly ONE `:root` block (the LIGHT base, 56 tokens). The DARK overrides use `:root[data-theme="dark"]`, `.dark`, and `@media(...) :root:not([data-theme="light"])` selectors, all of which the guard regex deliberately ignores — so the canonical token set stays clean while dark remains first-class in the browser.

4. Files created

- `design-systems/helix-ota/manifest.json` — od-design-system-project/v1, source bundled, declares files. `{design,tokens,tailwind}`.
- `design-systems/helix-ota/tokens.css` — 56-token `:root` LIGHT set + DARK overrides (3 toggle mechanisms).
- `design-systems/helix-ota/tailwind-v4.css` — Tailwind v4 @theme mapping, byte-exact with the renderer, compatible with ota-manager's Tailwind 4.
- `design-systems/helix-ota/DESIGN.md` — brand rationale + per-color provenance + §11.4.44 revision header (9 H2 sections).
- `docs/qa/20260709-helix-brand-tokens/EVIDENCE.md` — this file.

Note on scope: the package is intentionally a valid **prose + tokens + tailwind** Design System Project (it passes `check-design-system-manifests.ts`). It does NOT declare usage / `componentsManifest` / `preview` / `sourceFiles`, so it is not a “migrated/rich” package under

`check-design-system-package-quality.ts` and does not require `components.html` + preview pages — those are outside this task's scope (the task requested `manifest` + `tokens.css` + `tailwind-v4.css` + `DESIGN.md` only).

5. Flagged / UNCONFIRMED

- **UNCONFIRMED brand accent anchor:** no Helix-OTA-specific logo or brand asset exists in the repo. `helix_track/assets/HelixTrack-Logo.svg` belongs to a SIBLING project (`helix_track`), not `helix_ota`; anchoring the accent to it would be a guess (§11.4.6), so it was NOT used. The brand blue is taken solely from `clients/ota-manager/src/index.css --primary`. If a canonical Helix OTA logo is later provided, re-verify `--accent` against it.
- **Dark --danger contrast:** dark `--destructive` is `#7f1d1d` (a dark red used by `shadcn` as a fill behind a light foreground, not as body text). Faithful to the canonical source; if used as a foreground/icon color on the dark surface it should be paired with `--accent-on-style` light text, per the `shadcn` convention.